

Credit Card Customer Churn

From retrospective predictor to deployable early-warning system

Alexis

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FinTech — Prof. D. Marazzina
Politecnico di Milano

Agenda

1. Problem framing & data
2. The **FULL vs PROACTIVE** methodological choice
3. Pipeline & imbalance handling
4. Random Forest & MLP — training protocol
5. Results: ROC, PR & generalization
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8. **Probability calibration**
9. Business threshold F_2 tuning
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Problem framing

Problem & dataset

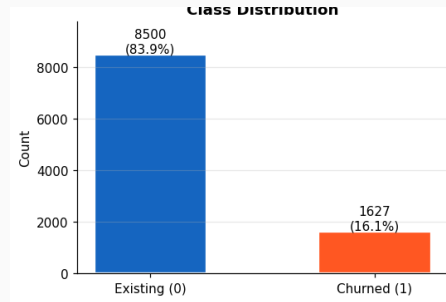
Setting. Identify customers likely to cancel their credit card *before* they leave so that account managers can intervene.

Three structural difficulties:

- **Class imbalance.** Only 16.1% churners.
- **Asymmetric cost.** Missed churn $\approx 10\times$ wasted call.
- **Predictor / proactive trap.** Most predictive features are *lagging symptoms*, not leading indicators.

Data hygiene. Drop Naive_Bayes_* (target leakage) and CLIENTNUM.

Engineer Balance_to_Limit_Ratio
and Contacts_per_Inactive_Month.



$n = 10,127$ customers, 20 features.
Stratified 80/20 split, scaler fit on train only.

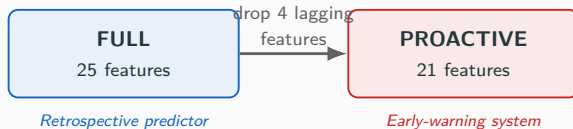
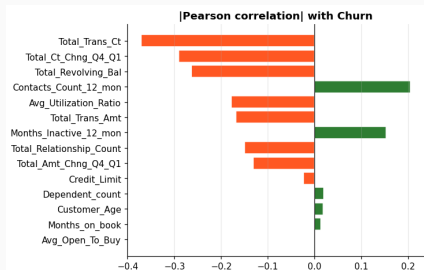
Predictor or proactive? — the central trade-off

The four most correlated features with churn are transactional aggregates measured *over the same 12-month window as the label*:

Total_Trans_Ct, Total_Ct_Chng_Q4_Q1,

Total_Trans_Amt, Total_Amt_Chng_Q4_Q1.

These are **lagging indicators**, not data leakage in the statistical sense.



Decision. Train every model on **both** configurations to *quantify* the cost of removing the lagging signal.

Why it matters. Once transaction volume drops, the customer is already disengaged.

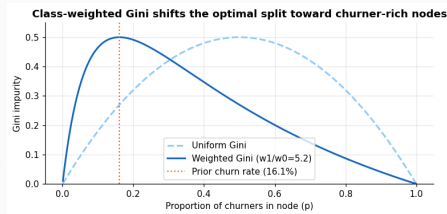
Methodology

Pipeline & imbalance handling

Mirrored pipeline. Stratified 80/20 split (preserving the 16.1% prior) → StandardScaler fit **on train only** → class weights $w_1/w_0 = 5.21$ injected into RF Gini and MLP loss → GridSearchCV with **scoring = average_precision** (not accuracy) → decision threshold tuned on **inner validation** predictions, never on test.

Why AP, not accuracy? The majority-class baseline already reaches 83.9% accuracy. *Average Precision* ignores the dominant true-negative mass and ranks the minority-class signal.

Sanity check (PROACTIVE). Tuned RF/MLP reach $AP \approx 0.56\text{--}0.58$ vs prior 0.160 — a $\sim \times 3.5$ **lift over chance**.



The class-weighted Gini peaks at the prior (16.1%): splits become reward-bearing precisely where churners concentrate, instead of at the symmetric 50% point.

Random Forest

- **Three-step protocol:**
 - B1: baseline, no class weights.
 - B2: `balanced_subsample`.
 - Tuned: GridSearchCV over depth, $n_estimators$, leaf size.
- `scoring = average_precision`, 5-fold stratified CV.
- PROACTIVE: regularized grid ($\max_depth \leq 20$, $\min_samples_leaf \geq 5$).

Multi-Layer Perceptron

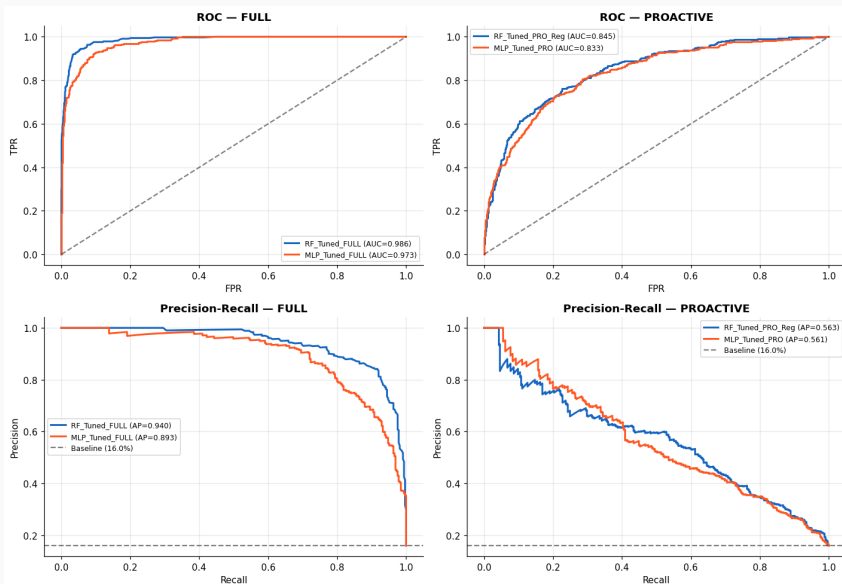
- Dense $\rightarrow$ BatchNorm $\rightarrow$ ReLU $\rightarrow$ Dropout stack.
- Best config (PROACTIVE):
 - $\text{hidden}=(128,64)$, $\text{dropout}=0.4$, $\text{lr}=10^{-3}$, $\text{lr}=5 \cdot 10^{-4}$
- EarlyStopping (patience 15) + ReduceLROnPlateau.
- Class weights injected directly into the loss.
- 5 hand-picked configs, inner validation split.

Anti-leakage discipline

Scaler fit on training fold only; hyperparameters chosen by CV on the *training* set; F_2 threshold optimized on **inner validation** predictions (never on test); test set touched *once* for final reporting.

Results

Performance — ROC and Precision-Recall



Headline metrics — RF vs MLP across configurations

Model	AP	AUC	Recall	Precision	F1
RF_Tuned_FULL	0.9403	0.9860	0.763	0.925	0.836
MLP_Tuned_FULL	0.8864	0.9707	0.883	0.698	0.780
RF_Tuned_PRO	0.5625	0.8446	0.625	0.505	0.559
MLP_Tuned_PRO	0.5671	0.8345	0.745	0.393	0.514

Three takeaways

1. **The configuration dominates the model family** — $\Delta\text{AP} \approx 0.38$ FULL $\rightarrow$ PRO, far larger than any RF/MLP gap.
2. On FULL the RF leads by ~ 0.05 AP; on PROACTIVE the gap is only 0.005 —deep inside bootstrap noise.
3. MLP *systematically* buys recall by trading precision (BatchNorm + Dropout regularizes the noisy proactive signal better).

Feature selection & interpretability

Feature selection — RFECV with the 1-sigma parsimony rule

Method. RFECV with a balanced-subsample RF, **5-fold** stratified CV, scoring = AP. Rather than picking the absolute peak, we apply **Occam's razor** (Hastie, Tibshirani & Friedman, ESL §7.10):

$$k^* = \min\{k : \overline{AP}_{CV}(k) \geq \overline{AP}_{CV}(k_{\text{peak}}) - \sigma(k_{\text{peak}})\}$$

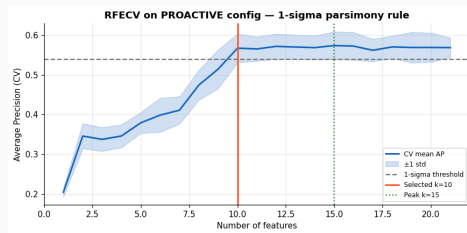
Peak : $k = 15$, $AP = 0.5740$

1σ threshold : $AP \geq 0.5389$

Selected : $k^* = 10$

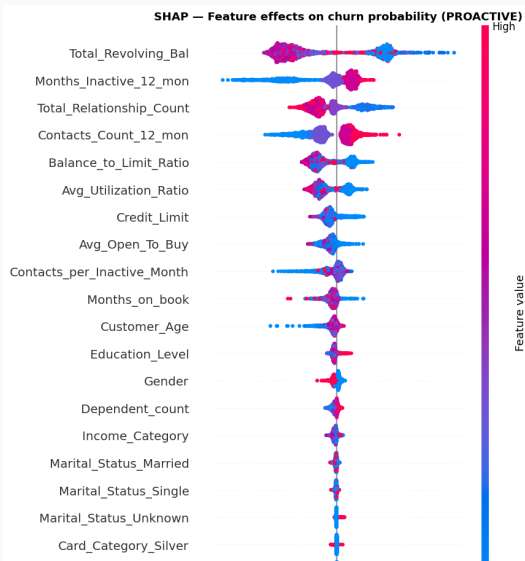
Selected features (by importance):

Total_Revolving_Bal, Balance_to_Limit_Ratio,
Avg_Utilization_Ratio, Avg_Open_To_Buy, Credit_Limit,
Customer_Age, Months_on_book,
Months_Inactive_12_mon, Total_Relationship_Count,
Contacts_Count_12_mon.



The plateau between $k = 10$ and $k = 15$ is structural: half of the PROACTIVE features carry essentially redundant information.

SHAP — what drives churn in the deployable model



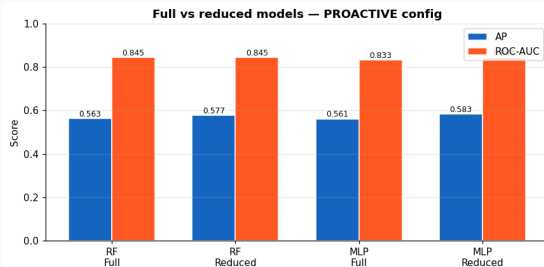
Reading the plot. Each dot is one test customer; horizontal position = effect on log-odds; colour = feature value.

Top drivers (PROACTIVE).

- Total_Revolving_Bal
low (blue) ↑ churn — no longer carries balance.
- Months_Inactive_12_mon
high (red) ↑ churn — prolonged inactivity.
- Total_Relationship_Count
low (blue) ↑ churn — single-product user.
- Contacts_Count_12_mon
high (red) ↑ churn — unresolved frustration.

Cross-check. MLP permutation importance confirms the *same cluster* of top drivers.

Impact of feature selection (PROACTIVE config)



Model (21 → 10 features)	AP	Recall
RF_PRO (21 ft)	0.5625	0.625
RF Reduced (10 ft)	0.5811	0.628
MLP_PRO (21 ft)	0.5671	0.745
MLP Reduced (10 ft)	0.5838	0.754

Both models improve under parsimony:

- RF: +0.019 AP (regularization through input compression).
- MLP: +0.017 AP.
- RF 0.581, MLP 0.584 —**statistically tied, best of the study**.
- ROC-AUC essentially unchanged $\Rightarrow$ no ranking information lost.

Operational win. Half the features to monitor, lower drift surface, identical performance.

Business deployment

Probability calibration — making scores business-interpretable

Why calibrate?

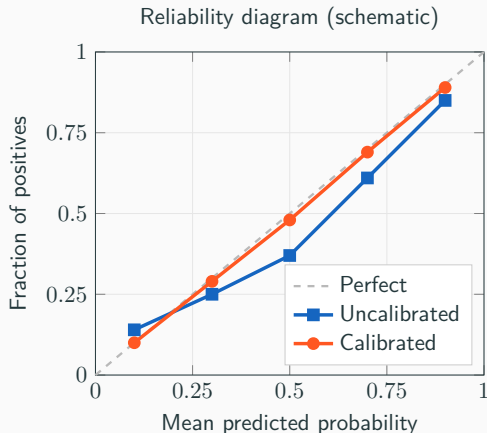
Raw RF scores cluster near 0 and 1 (overconfident distribution). `CalibratedClassifierCV` maps them to genuine probabilities, making τ interpretable as a true churn risk.

Model	AP	AUC	Brier
Uncalibrated	0.581	0.846	0.1140
Calibrated	0.575	0.843	0.0971

Brier ↓ +0.017 | **AUC** Δ = −0.002.

💡 Key insight

Calibration corrects *magnitudes*, not ranking. AUC is essentially preserved; Brier drops by 15%.



Business threshold optimization — F_2 over default 0.5

Why $\tau = 0.5$ is wrong. 0/1 loss ignores the $\sim 1 : 10$ cost asymmetry. The natural deployment objective is:

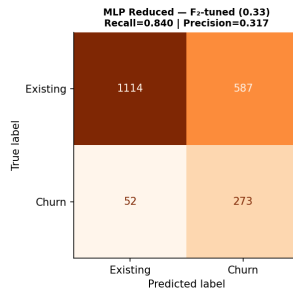
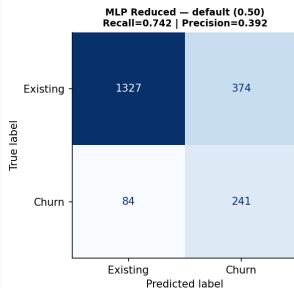
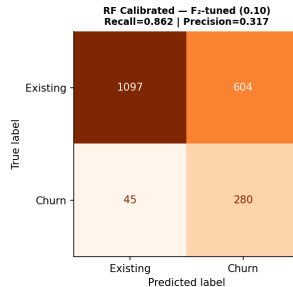
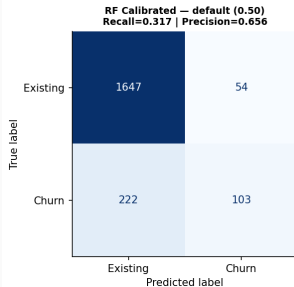
$$F_\beta = (1 + \beta^2) \frac{\text{Precision} \cdot \text{Recall}}{\beta^2 \text{Precision} + \text{Recall}}, \quad \beta = 2 \implies \text{recall weighted } 4 \times \text{precision}.$$

Discipline. τ^* chosen on **inner validation** predictions (both RF calibrated and MLP) — *never* on test.

Model	τ	Recall	Precision	F_2	Caught	Alerts
RF cal. default	0.50	0.317	0.656	0.354	103/325	157
RF cal. F_2-tuned	0.098	0.862	0.317	0.641	280/325	884
MLP Reduced default	0.50	0.754	0.391	0.636	245/325	626
MLP Reduced F_2 -tuned	0.330	0.858	0.316	0.639	279/325	882

Result. RF F_2 -tuned catches **+177 churners** ($103 \rightarrow 280$) over the conservative default. **Caveat:** $\tau = 0.098$ flags 44% of the base (884/2,026 alerts) — use the cost table to pick the right τ for your FN:FP ratio.

Confusion matrices — visualising the trade-off



Expected-cost analysis — beyond the F_2 proxy

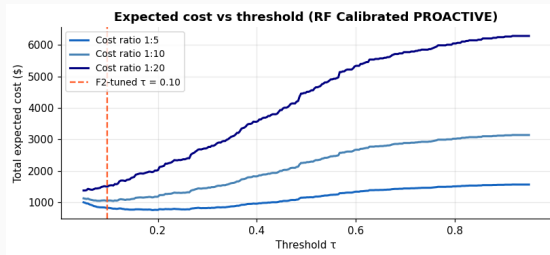
Total expected cost on the test set, RF Calibrated PROACTIVE:

$$C(\tau) = c_{FN} \cdot FN(\tau) + c_{FP} \cdot FP(\tau)$$

Ratio	τ^*	Recall at τ^*
1:1	0.462	0.391
1:5	0.188	0.754
1:10	0.110	0.849
1:20	0.050	0.923

Diagnostic.

- The F_2 -optimal $\tau = 0.098$ sits in the **1:10 regime** ($\tau^* = 0.110$, recall ≈ 0.85).
- At 1:5 ($\tau \approx 0.19$, recall 0.75): more conservative and operationally safer.
- Choose τ from this table, not from F_2 alone.



Conclusion

Deployment recommendation & caveats

Production model

RF_PRO_Reduced_Calibrated

10 leading-indicator features.

- **True risk:** Calibrated τ is a genuine probability.
- **High recall ($\tau \approx 0.11$): 85% of churners caught (1:10 ratio).**
- **Balanced ($\tau \approx 0.19$):** 75% caught (1:5 ratio).
- **Avoid F_2 -optimal:** $\tau = 0.098$ flags 44% of the base.
- **Operations:** SHAP interpretable; requires monthly retraining.

Honest caveats

- **Overfitting:** RF in-sample gap +0.33, but OOB→test drops to +0.014.
- **Parity:** RF vs MLP performance sits within bootstrap noise.
- **Validation:** Single time-split; no concept-drift validation yet.
- **Features:** Avg_Open_To_Buy is a linear dependency (harmless for trees).